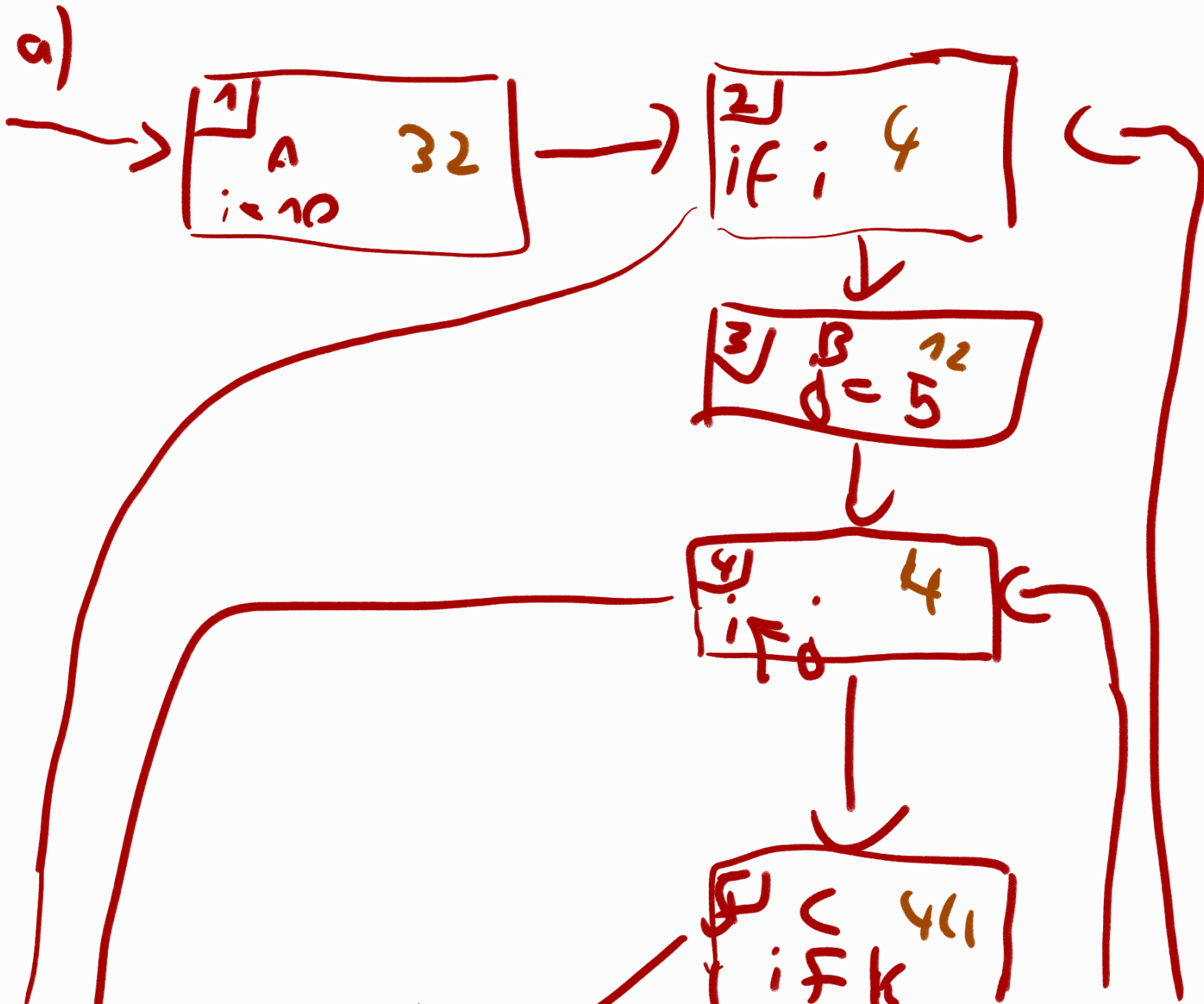


Exercise 4

For scheduling in hard real-time systems it is mandatory to verify that no task deadlines are missed during run-time. In order to guarantee such safety requirements you generally have to perform a worst-case execution time (WCET) analysis. In this exercise you have to determine the WCET of the following code snippet which is presented in a C-like format:

```
// the variables i, j and k are of type int
entry:
  someCode_A; // 30 cycles
  i = 10; // 2 cycles
label_1:
  if (i == 0) goto label_0; // 4 cycles
  someCode_B; // 10 cycles
  j = 5; // 2 cycles
label_3:
  if (j == 0) goto label_2; // 4 cycles
  someCode_C; // 40 cycles
  if (k == 0) goto label_4; // 4 cycles
  someCode_D; // 50 cycles
  goto label_5; // 10 cycles
label_4:
  someCode_E; // 20 cycles
label_5:
  j = j - 1; // 2 cycles
  goto label_3; // 3 cycles
label_2:
  i = i - 1; // 2 cycles
  goto label_1; // 3 cycles
label_0:
```

- Draw the control flow graph (CFG) of the program, such that nodes correspond to basic blocks and edges show successor relationships.
- Formulate an *Integer Linear Programming (ILP)* problem for the WCET analysis of the control flow graph. The program contains two loops. The cost function, and therefore worst-case execution time, of your ILP problem will be unbounded if you do not specify an upper limit for loop iterations. Give additional constraints so that the body of the "outer" loop is executed at most $i = 10$ times, and the "inner" loop body is executed at most $j = 5$ times in one pass of the outer loop.



7
E 20

610 S₃
gate

810 S
gate

410 S
gate

101
table 1

$$b) \max x \quad 32x_1 + 4x_2 + 12x_3 + 4x_4 \\ + 44x_5 + 53x_6 + 20x_7 \\ x_1 = x_{i1} \quad x_{i1} = 1 \quad 5x_8 + 5x_9 \\ x_3 = x_{2,3} + 0x_{10}$$

$$x_1 = x_{1,2} \quad x_3 = x_{3,4}$$

$$x_4 = x_{3,4} + x_{8,4}$$

$$x_2 = x_{1,2} + x_{9,2} \quad x_4 = x_{4,5} + x_{4,7}$$

$$x_2 = x_{2,3} + x_{2,10}$$

$$x_5 =$$

$$x_5 = x_{4,5}$$

$$x_6 = x_{5,6}$$

$$x_6 = x_{6,8}$$

$$x_5 = x_{5,6} + x_{5,7} \quad x_7 = x_{6,8} + 6x_{7,8} \\ x_7 = x_{8,4}$$

$$x_7 = x_{4,7}$$

$$x_9 = x_{9,2}$$

$$\forall i: x_i \geq 0$$

$$x_{10} = x_{2,10}$$

$$\forall i, j: x_{ij} \geq 0$$

$$x_{10} = x_{10, out}$$

$$x_{out} = 1$$

$$\text{inner: } x_8 \leq 50$$

$$\text{outer: } x_9 \leq 10$$

$$x_{8,4} \leq 50$$

$$x_{9,2} \leq 10$$

Exercise 3

Assume the control flow graph of a program shown in Fig. 1. Perform a value analysis for x and y by

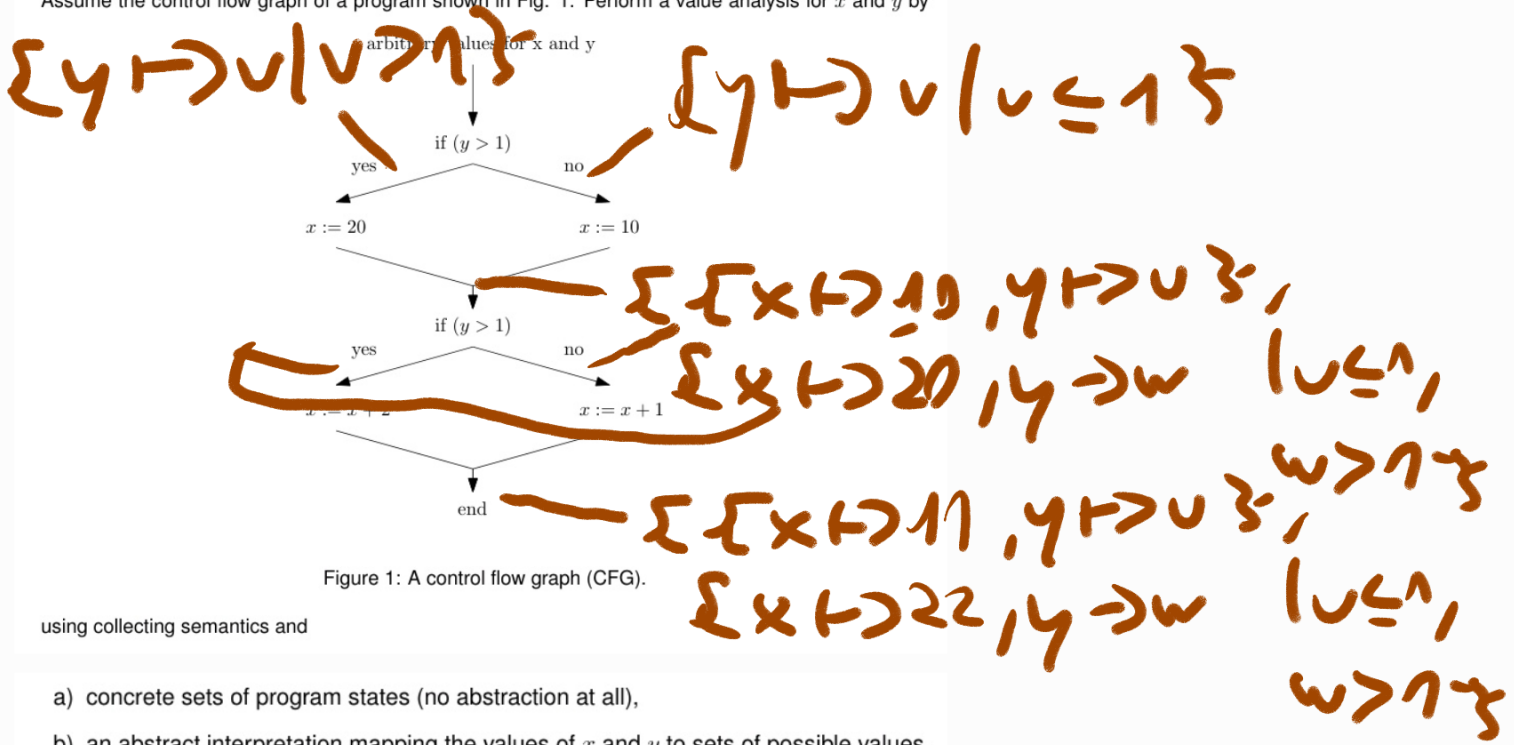


Figure 1: A control flow graph (CFG).

using collecting semantics and

- concrete sets of program states (no abstraction at all),
- an abstract interpretation mapping the values of x and y to sets of possible values,
- an abstract interpretation mapping the values of x and y to intervals.
- an abstract interpretation mapping the values of x and y to subsets of $\{+, -, 0\}$.

Compare the accuracy of the different approaches.

→ concrete
* 14 lost

$\{x \mapsto \{10, 20\}\}$

$y \mapsto \top$

$\{x \mapsto \{11, 21, 22\}, y \mapsto \top\}$

→ lost

$\{x \mapsto [10, 20]\}$

$y \mapsto [-\infty, \infty]$

$\{x \mapsto [11, 22]\}$

$y \mapsto [-\infty, \infty]$

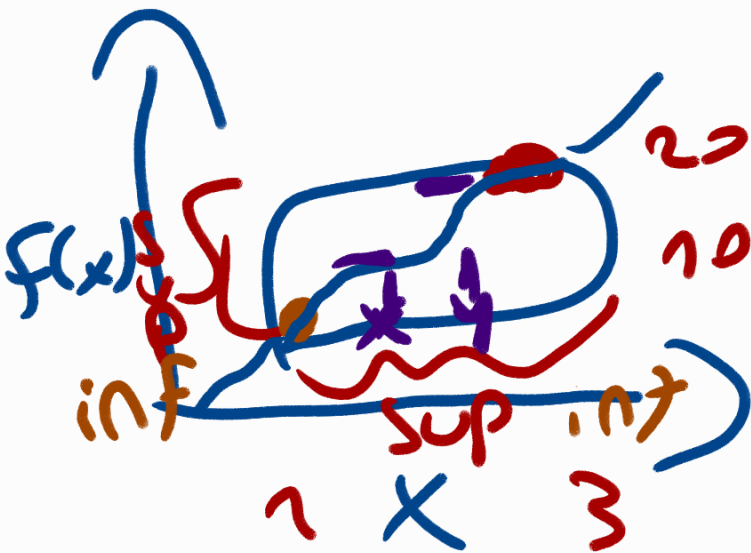
* 5/14
will be

$\{x \mapsto f(x)\}$ in a row
 $y \mapsto \{0, +, -\}$
 $\{x \mapsto f(x)\}$
 $y \mapsto \{0, +, -\}$

Exercise 2

Prove the following lemma of the lecture: If $f : A \rightarrow A$ is a function that is either upward or downward continuous, then f is monotonically increasing.

$$x \leq y \rightarrow f(x) \leq f(y)$$



Let x, y be elements of $(A, <)$ and f be upward or down continuous.

Case 1: f upward continuous

$$f(x) \leq \sup\{f(z) \mid 1 \leq z \leq x\}$$

$$\begin{aligned}
 \max & \quad = \sup \{ f(\Sigma x, y) \} \\
 & = f(\sup \{ \Sigma x, y \}) \\
 & = f(y)
 \end{aligned}$$

$$\begin{aligned}
 f(y) & \geq \inf \{ f(x), f(y) \} \\
 & = \inf \{ f(\Sigma x, y) \} \\
 & = f(\inf \{ \Sigma x, y \}) \\
 & = f(x)
 \end{aligned}$$

Exercise 1

Prove (by induction) the following lemma of the lecture: All finite lattices are complete.

Let $(M, <)$ be a finite lattice
Induction Basis w.l.o.g. use $f, \sup$
Let $(M, <)$ w. $M = \{x\}$ be $|M| = 1$
 $\rightarrow$ there's only one non-empty
subset: $\sup(\{x\})$
 $= x$

Induction step:

Let $(M, <)$ w. $M = M' \cup \{x\}$
be a f. lattice
- M' is a complete lattice
due to induction hyp.

CUZ

Let $\bar{S}$ be a non-empty subset
of M , then

Case 1: $x \notin S$

- Then S is a $\subseteq$ subset of
 M and by induction h.

$\sup(S)$ exists, CUZ subset
of M

(- $\langle x \rangle$ is complete lth of
due to induction basis)

Case 1: $x \in S$

Case 1.1 $|S| = 1$:

S only contains x , it
holds $\sup(S) = x$

Case 1.2 $|S| > 1$

$$S = S' \cup \{x\}$$

- S' is subset of M'

and by the induction

h. $\sup(S')$ exists

~ $\sup(S)$ can be constructed
by $\sup(\sup(S'), x)$

